1.2.2 Container as a Service (CaaS)

Using the *Container as a Service* (CaaS) model, consumers cannot control primitive virtualization resources. Instead, they provision and manage containers. The cloud provider takes care of provisioning the underlying resources that fulfill the needs of those containers, such as by starting new virtual machines and configuring networks to make them accessible through the internet. Docker Swarm, Apache Mesos, and Kubernetes are examples of tools used to build container platforms. All major cloud providers offer a managed Kubernetes service, which has become the de facto technology for CaaS offerings: Amazon Elastic Kubernetes Service (EKS), Azure Kubernetes Service (AKS), Google Kubernetes Engine (GKE), Alibaba Container Service for Kubernetes (ACK), and DigitalOcean Kubernetes.

1.2.3 Platform as a Service (PaaS)

In the *Platform as a Service* (PaaS) model, the platform provides infrastructure, tools, and APIs that developers can use to build and deploy applications. For example, as a developer, you can build a Java application, package it as a Java Archive (JAR) file, and then deploy it to a platform working according to the PaaS model. The platform provides the Java runtime and other required middleware, and it can also offer extra services like databases or messaging systems. Examples of PaaS offerings are Cloud Foundry, Heroku, AWS Elastic Beanstalk, Azure App Service, Google App Engine, Alibaba Web App Service, and DigitalOcean App Platform. In the past few years, vendors have been converging on Kubernetes for building a new PaaS experience for developers and operators. Examples of this new generation of services are VMware Tanzu Application Platform and RedHat OpenShift.